

On Four Definitions of Dimension

In lecture on Friday, March 2, 2018, we saw four definitions of dimension for affine algebraic varieties, and an argument toward their equivalence. Here we fill the gaps. I thank Arpan Pal, Byeongsu Yu, and Nida Obatake for pointing out the gaps!

Remark 1 *I have intentionally diverged from Shafarevich's treatment for two reasons: (1) Having several distinct definitions of dimensions at one's finger-tips makes it much easier to prove the numerous properties of dimension. (2) One learns more deeply by seeing more than one point of view. $\diamond$*

In what follows, we let K denote an algebraically closed field and $X \subseteq K^n$ an affine variety (i.e., an irreducible Zariski-closed subset).

Definition 0 *Let $\dim X$ be -1 (resp. 0) if X is empty (resp. X is a point). Otherwise, let $\dim X$ be the maximal d such that there is a cardinality d subset $\{i_1, \dots, i_d\} \subseteq \{1, \dots, n\}$ making the coordinate projection $\pi : K^n \rightarrow K^d$ defined by $\pi(x_1, \dots, x_n) := (x_{i_1}, \dots, x_{i_d})$ have $\pi(X)$ has non-empty Zariski-interior in K^d . $\diamond$*

Definition 1 *Let $\dim X$ be -1 if X is empty. Otherwise, let $\dim X$ be the transcendence degree (over K) of the function field $K(X)$. $\diamond$*

Definition 2 *Let $\dim X$ be -1 (resp. 0) if X is empty (resp. X is a point). Otherwise, let $\dim X$ be the maximal d such that there is a descending chain of non-empty irreducible closed sets $X_1, \dots, X_d$ satisfying*

$$X \supsetneq X_1 \supsetneq \dots \supsetneq X_d$$

and X_d a point. $\diamond$

Definition 3 *Let $\dim X$ be -1 (resp. 0) if X is empty (resp. X is a point). Otherwise, let $\dim X$ be the unique d such that there is a finite regular map $f : X \rightarrow K^d$. $\diamond$*

Each definition has its strengths: Definition 0 is one of the easiest to use algorithmically, and can be extended almost effortlessly to complex analytic sets, semi-algebraic subsets over real-closed fields, and sub-analytic sets. Definition 1 is clearly a birational invariant. Definition 2 uses nothing more than the Zariski topology. Definition 3 leads us to consider an important class of regular maps useful in other contexts.

A central result behind the equivalence of these definitions is the notion of *Noether Normalization*. A refined affine version, implicit in Shafarevich's development in Section 5 of Chapter 1 of our text, is the following.

Noether Normalization (Affine Version) *For any infinite irreducible Zariski-closed X in K^n , there is a positive integer d and linear forms $L_1, \dots, L_d \in K[x_1, \dots, x_n]$ such that the linear projection $L : X \rightarrow K^d$ defined by $L(x) := (L_1(x), \dots, L_d(x))$ is a finite regular map. $\blacksquare$*

Remark 4 *One can in fact show that a generic linear map from X to $K^{\dim X}$ will be finite. This has connections to Bertini's Theorem and Morse Theory that we'll touch upon later. $\diamond$*

We now prove the following theorem:

Theorem 5 *The preceding four definitions are well-defined and identical for any irreducible affine X .*

Proof: Definitions 0–2 are clearly well-defined. That the finite regular map asserted in Definition 3 exists follows from our statement of Noether Normalization. That Definition 3 leads to a *uniquely* defined quantity will then follow from the equivalence of Definitions 1 and 3, which we prove now:

Definition 1 = Definition 3: The case where X is empty is trivial. If X is a point then $K(X) = K$, which clearly has transcendence degree 0 over K . So we may assume X is infinite, and that the transcendence degree of $K(X)$ over K is d .

By Affine Noether Normalization, for some positive integer δ , there must be an embedding of $K[u_1, \dots, u_\delta]$ into the coordinate ring $K[X]$ such that $K[X]$ is an integral extension of $K[u_1, \dots, u_\delta]$ (for some indeterminates $u_1, \dots, u_\delta$ algebraically independent over K). Taking quotient fields, it then follows that $K(X)$ is integral over $K(x_1, \dots, x_\delta)$. (This is a basic commutative algebra fact: See, e.g., Proposition 4.6, Page 61 of Commutative Algebra, by Atiyah and MacDonald.) In particular, $K(X)$ is algebraic over $K(u_1, \dots, u_\delta)$. In other words, $K(X)$ has transcendence degree δ over K , and thus $d = \delta$, and we are done. ■

Definition 1 = Definition 2: Similar to our proof that Definitions 1 and 3 agree, we may assume X is infinite and that $K(X)$ has positive transcendence degree d_1 . Let d_2 be the dimension of X with respect to Definition 2. We will show that $d_2 \geq d_1$ and then $d_1 \geq d_2$, thus proving $d_1 = d_2$.

$[d_2 \geq d_1]$ Since we've already proved that Definitions 1 and 3 are equivalent, we may assume that there is a finite regular map $f : X \rightarrow K^{d_1}$. This in turn implies that $f^* : K[u_1, \dots, u_{d_1}] \rightarrow K[X]$ is injective and $K[X]$ is integral over $f^*(K[u_1, \dots, u_{d_1}])$. Since $X \subseteq K^n$ we may assume $K[X] = K[x_1, \dots, x_n]/I$ for some ideal I , let $f_i := f^*(u_i)$ for all i , and regard the f_i as polynomials in $x_1, \dots, x_n$. Note in particular that the f_i are algebraically independent in $K(X)$ since f^* is an embedding of K -algebras.

Now let $W_1, \dots, W_r$ be the irreducible components of the closed subset W of X defined by the equation $f_{d_1} = 0$. Assume, to later derive a contradiction, that $f_1, \dots, f_{d_1-1}$ are algebraically dependent on each W_i , i.e., that a non-trivial polynomial equation $P_i(f_1, \dots, f_{d_1-1}) = 0$ holds in each ring $K[W_i] = K[X]/I(W_i)$. Then $P := \prod_{i=1}^r P_i$ is identically 0 on W . In other words, since f^* is injective, $P(u_1, \dots, u_{d_1-1}) = 0$ in $K[u_1, \dots, u_{d_1}]$ for some non-constant polynomial P . This contradicts the algebraic independence of $u_1, \dots, u_{d_1-1}$, and thus we may assume that $f_1, \dots, f_{d_1-1}$ are algebraically independent on some W_i , which we will call X_1 . Note in particular that $K(X_1)$ has transcendence degree $d_1 - 1$ over K .

Repeating the preceding argument, we obtain a descending chain of irreducible closed subsets $X \supsetneq X_1 \supsetneq \dots \supsetneq X_{d_1}$, since we can perform the preceding procedure as long as the underlying transcendence degree is positive. Clearly then, a *maximal* descending chain from X must have length at least d_1 and thus $d_2 \geq d_1$. ■

$[d_1 \geq d_2]$ Suppose on the other hand that we start with a chain of length d_2 : $X \supsetneq X_1 \supsetneq \dots \supsetneq X_{d_2}$, with X_{d_2} a point. Note that the corresponding ideals in $K[X]$ form a strictly increasing chain:

$$\langle 0 \rangle = I(X) \subsetneq I(X_1) \subsetneq \dots \subsetneq I(X_{d_2}) \neq K[X].$$

We can then clearly find, for each $i \geq 1$, a polynomial $f_i \in K[X]$ which vanishes identically on X_i but not on X_{i-1} (setting $X_0 := X$). If $f_1, \dots, f_{d_2}$ are algebraically independent then we'll have shown that the transcendence degree of $K(X)$ is at least d_2 , i.e., we'll have at last proved $d_1 \geq d_2$. Toward this end, let us successively show that $f_i, \dots, f_{d_2}$ are algebraically independent in $K(X_{i-1})$, as i decreases from d_2 to 1.

First note that, since K is algebraically closed, f_{d_2} algebraic over K implies that $f_{d_2} \in K$. Since f_{d_2} is non-constant, f_{d_2} is indeed not in K , and thus can not be a root of a polynomial in $K(X_{d_2-1})$. Proceeding inductively, now assume $f_{i+1}, \dots, f_{d_2}$ are algebraically independent in $K(X_i)$. To derive a contradiction, assume that $f_i, \dots, f_{d_2}$ are algebraically dependent in $K(X_{i-1})$. Then an identity of the form $G(f_i(z), \dots, f_{d_2}(z)) = 0$, for some non-constant $G \in K[u_i, \dots, u_{d_2}]$, must hold for all $z \in X_{i-1}$. Since X_{i-1} is irreducible, X_{i-1} must be contained in the zero set of some irreducible factor of G . So we may clearly assume that G is irreducible. Note also that, by construction, f_i does not vanish on all of X_{i-1} . So we may assume $G(u_i, \dots, u_{d_2}) = \sum_{j=0}^m G_j u_i^j$ with $G_0, \dots, G_m \in K[u_{i+1}, \dots, u_{d_2}]$ and $G_0 \notin K$. But then $G_0(f_{i+1}, \dots, f_{d_2})$ vanishes identically on X_i , contradicting the algebraic independence of $f_2, \dots, f_{d_2}$ in $K(X_i)$! ■

Remark 6 *For the sake of posting this draft quickly, I'll leave the proof that Definition 0 agrees with Definitions 1–3 as an exercise. $\diamond$*

Now that we know that our four roads lead to the same place, we can easily extend the definition of dimension to more general classes of sets.

Definition 7 *If X is any non-empty algebraic set (reducible or irreducible), then we define $\dim X$ to be the maximum of $\dim X'$ as X' ranges over all irreducible components of X . $\diamond$*

The preceding notion is clearly well-defined. The next three extensions actually require a little checking:

Definition 8 *If X is any non-empty quasi-affine subset of the form $Y \setminus Z$, with Y and Z Zariski-closed, and $Y \neq Y \cap Z$, then we define $\dim X := \dim Y$. $\diamond$*

Definition 9 *If $X \subseteq K^n$ is any non-empty constructible subset of the form $X_1 \cup \cdots \cup X_r$ with each X_i quasi-affine, then $\dim X := \max_{i \in \{1, \dots, r\}} \dim X_i$. $\diamond$*

Definition 10 *If $X \subseteq \mathbb{P}_K^n$ is any non-empty constructible subset, and we coordinatize via $\mathbb{P}_K^n = \{[x_0 : \cdots : x_n]\}$, then $\dim X := \max_{i \in \{0, \dots, n\}} \dim(X \cap \{x_i \neq 0\})$. $\diamond$*

We leave the well-definedness of the last 3 definitions as an exercise.